

Samir Pahari

2026 Outstanding Geoscience Graduate Student

Graduate Research and Teaching Assistant, Landscape Geodynamics (LeGo) Lab
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Overview

MS in Geoscience candidate at Western Kentucky University (GPA 4.0) with professional experience spanning graduate teaching/research and municipal stormwater GIS. Specializes in GIS, remote sensing, and spatial analysis for environmental and hazard applications, including RUSLE-based soil loss assessment and deep-learning flood susceptibility mapping. Proficient in the ESRI ecosystem (ArcGIS Pro/Enterprise/Online, Field Maps, Experience Builder) with Python (ArcPy), SQL, and GNSS field data collection/QA/QC, and known for clear communication and hands-on instruction.

Education

Fully funded Ph.D. in Earth Science; Starting in Fall 2026

- School of Earth Sciences, The Ohio State University, Columbus, Ohio

M.S. in Geoscience (August 2024- Present); Expected graduation in Spring 2026

- Western Kentucky University Bowling Green, Kentucky
- Grade: 4.0

B.S. in Geomatics Engineering (November 2018 – July 2023)

- IOE Pashchimanchal Campus, Tribhuvan University, Pokhara, Nepal
- Percentage: 77.93%

Peer-reviewed Publication

- **Pahari, S.** and Gani, N.D., 2025. Soil Loss Analysis of North Fork Kentucky River Basin Utilizing the Revised Universal Soil Loss Equation (RUSLE). Journal of the Kentucky Academy of Science, 86(1), pp.1-15. <https://doi.org/10.3101/KYAC-86-01-01>

Manuscript in Preparation

- **Pahari, S.** Gani, N.D, Singh, M., Aham, V. Geology-aware, patch-based deep learning for flood susceptibility mapping in a confined Appalachian watershed, eastern Kentucky, USA. (in review, Natural Hazards)

Conference Abstract & Presentations

- **Pahari, S.** and Gani N.D. 2026. High-resolution machine learning-based flood susceptibility Mapping in the North Fork Kentucky River basin, Eastern Kentucky. 56th Annual WKU Student Scholar Showcase, April 11 (in review).
- **Pahari, S.** and Gani N.D. 2025. Advancing Flood Susceptibility Prediction with Deep Learning in North Fork Kentucky River Basin, Eastern Kentucky. AGU25. <https://agu.confex.com/agu/agu25/meetingapp.cgi/Paper/1902919>
- **Pahari, S.** and Gani N.D. 2025. Deep Learning for Improved Flood Susceptibility Prediction in Eastern Kentucky's North Fork River Basin. Kentucky GIS Conference Florence, KY.
- **Pahari, S.** and Gani N.D. 2025. Leveraging Deep Learning and Remote Sensing-based Geomorphology for Flood Susceptibility Mapping in The Appalachian Mountain Regions of KY. 55th Annual WKU Student Scholar Showcase, April 5.
- **Pahari, S.** and Gani N.D. 2025. Leveraging Deep Learning and Remote Sensing-based Geomorphology for Flood Susceptibility Mapping in The Appalachian Mountain Regions of KY. 4th Annual Data Science Day, Applied Center for Data Science, April 8.
- Lowman, C., Hughes, M., Aham, V., A., **Pahari, S.**, Gani, M., and Gani, N.D., 2025. Geology in Flight: Unveiling WKU's Campus Karren Through Drone Technology. 55th Annual WKU Student Scholar Showcase, April 5.

Professional Work Experience

Graduate Teaching and Research Assistant (August 15, 2024 – Present)

Department of Earth, Environmental and Atmospheric Sciences, Western Kentucky University

- GEOL 113 (The Earth Lab) & GEOL 114 (Earth's Past & Future Lab)

- o Lead laboratory sessions for 20+ undergraduate students facilitating hands-on learning of key geological concepts and GIS applications.
- o Evaluate and grade student assignments and lab reports, providing constructive feedback to support student academic development and success.
- GEOL 111 (The Earth Online section) Assisted with grading, Q/A, communicating with students regularly.
- GEOL 350 (Mineralogy & Petrology) – Assisted with laboratory instruction, training students’ petrographic skills
- GEOL 380 (Intro Field Techniques) - Assisted with field trips, sample processing, and field equipment training.
- Trained multiple undergraduate students drone technology, 3D scanning, petrography for their research projects.
- M.S. thesis research - Conduct research on flood susceptibility mapping using deep learning techniques and ArcGIS pro, contributing to innovative solutions in environmental risk assessment.

Geographic Information Science Technician (May 12- August 8, 2025)

Warren County Stormwater, Bowling Green, KY

- Work as an intern for Geographic Information Science Technician, utilizing ArcGIS pro and SQL database to create feature classes representing storm water assets.
- Hands-on GNSS field work (R2 GNSS Receivers), cross-departmental rotations and post-collection QA/QC.
- Built domains and relationship classes to enable data collection, deployed to portal by publishing feature layers.
- Utilized ArcGIS field maps to refine form, including arcade to collect attribute collection and list to standardize the data inputs.
- Attended two public outreach events as part to actively engage in promoting sustainable stormwater management.

Geomatics Instructor (December 2023- June 2024)

Redcross Secondary School, Dailekh, Nepal

- Instructed diploma-level courses in geomatics engineering, covering essential subjects such as GIS, Remote Sensing, Photogrammetry, Geodesy, and Computer Programming.
- Led the Remote Sensing and GIS facility, ensuring the integration of advanced technologies into the curriculum.
- Enhanced student understanding and practical skills through hands-on projects using ArcGIS and remote sensing applications in geomatics.

Technical Proficiencies & Software Expertise

- **GIS Software:** ArcGIS, ArcGIS pro, ArcGIS Enterprise, ArcGIS online, ESRI Model Experience Builder, Survey Maps, QGIS, Google Earth Engine
- **Programming:** Python (ArcPy), Java Basics, R (Statistical Analysis), Arcade
- **Field Equipment:** Trimble Mobile Manager, R2 GNSS Receivers, UAV/ Drone Systems
- **Others:** Remote Sensing Applications, Machine Learning, Deep Learning, Drone Image Processing, Virtual Reality Sandbox, 3-D scanner.

Academic Projects

Comparative Analysis of Major Flood Events in Southeastern United States, December 2024

- Comprehensive Analysis of atmospheric parameters using ERA5 reanalysis dataset to investigate two major flood events of Nashville 2010 and Kentucky 2022.
- Developed time-series visualization and spatial analysis including multiple atmospheric parameters (temperature advection, wind, humidity, CAPE, CIN and IVT), using Python and GIS.

Form Factors and Diameter Height Modeling of Sal (Shorea robusta) in Nepal, August 2023

- Provided technical assistance for thesis project at the Institute of Forestry (IOF), Pokhara Campus, implementing advanced geospatial methodologies and analysis techniques.
- Specialized in applying Remote Sensing technologies and GIS tools for forest resource assessment, while utilizing programming skills to develop automated workflows and data processing solutions.
- Generated detailed distribution maps using ArcGIS Pro's advanced cartographic capabilities, incorporating multiple environmental variables and forest inventory data to create comprehensive spatial visualizations.

Optimizing Tomato Production with Precision Agriculture: A Water Variability Assessment Approach, May 2023

- Managed a broad set of images of tomato leaves under four different levels of field capacity (25%, 50%, 75%, and 100%), implementing systematic data collection and labelling protocols to ensure accurate training data.

- Successfully designed and trained a deep learning model on image classification using convolutional neural networks obtaining reliable accuracy classifying different field capacity levels by using leaf characteristics.
- Created and deployed a user-friendly web application by integrating trained model using Python backend frameworks combined with HTML and CSS for seamless, interactive user experiences and real-time predictions.

Grants for Research Presentation

- WKU Graduate School Student Travel Grant (\$500).
- Department of Earth, Environmental, and Atmospheric Sciences Travel Grant (\$500).

Conference Participation

- Attended Kentucky Stormwater Association (KSA) 2025 conference as a moderator of the program.
- Presented at Kentucky Associations of Mapping Professionals (KAMP) Conference Florence, September 24, 2025.
- Presented at American Geophysical Union (AGU) Conference, New Orleans, Louisiana, December 15-19, 2025.
- Presented and attended at the 55th Annual WKU Student Scholar Showcase, April 5, 2025.

Training & Certification

- Supervised Machine Learning: Regression and Classification, and Advanced Learning Algorithms, an online non-credit course authorized by DeepLearning.AI and Stanford University.
- Training on DGPS and Terrestrial LiDAR Scanner organized by Centre for Space Science and Geomatics Studies (CSSGS) and supported by University Grants Commission.
- Active and Passive Satellite Data Analysis Using Cloud Computing for Surface Water/ Flood Monitoring.
- A workshop on GNSS organized by Centre for Spatial Information Science, University of Tokyo (CSIS/UT) and International Committee on Global Navigation Satellite System (ICG).

Community/Civic Services & Outreach

- Departmental service: Supports program-level needs by helping catalog, clean, and modernize the department's sample collections. Actively contributes to recruiting and outreach efforts, including the Gatton and Honors College Research Fair, Geology Club activities, and other departmental recruiting events.
- Conference service: Served as a moderator at the KAMP Conference.
- Community outreach: Volunteered at the Warren County Public Library Summer Reading Kickoff, helping school-aged children learn about stormwater and watershed stewardship.
- Public engagement: Represented the Warren County Stormwater program as its mascot at the Duncan Hines Classic Car Cruise & Stroll Parade.
- K-12 education: Served as a guest science educator at Potter Gray Elementary School (Bowling Green), delivering an interactive geology and mineral identification lesson to a 2nd-grade class.

Current Professional Affiliations

- American Geophysical Union
- Kentucky Academy of Science
- Kentucky Academy of Mapping Professionals